MOTOROLA

M68EDITORM(D)

M6800EDITORM

RESIDENT EDITOR

REFERENCE MANUAL

silicium

MICROSYSTEMS

M6800EDITORM

RESIDENT EDITOR REFERENCE MANUAL

VERSION 3.0

TABLE OF CONTENTS

LIST OF ILLUSTRATIONS

LIST OF TABLES

CHAPTER 1

GENERAL INFORMATION

1.1 INTRODUCTION

Motorola's M6800EDITORM Resident Editor provides the users of Motorola development systems with powerful text editing capabilities. The Editor simplifies both the initial text and source program entry, as well as subsequent program modification.

The Editor permits many modes of operation. As a disk based editor, text may be entered either from a diskette file or terminal keyboard. The Editor will operate with any TTY compatible terminal. Use of the Editor in conjunction with the EXORterm 200/220 or EXORterm 150/EXORciser system allows the user to perform editing, employing specifically designed features of the EXORterm.

The Editor provides a standard set of editing functions, as found in most microprocessor editors:

- insert string/line
- change string
- list program
- delete string/line

In addition, the Editor allows more powerful editing functions, such as:

- multi-string replacement
- block moves
- verification of editing changes
- CRT oriented text modification
- editing ranges
- tabbing

1.2 MINIMUM CONFIGURATION FOR EDITOR OPERATION

The Editor requires as a minimum configuration the following:

- TTY compatible terminal or EXORterm 200 or EXORterm 220 or EXORterm 150/EXORciser system
- 32K Random Access Memory
- EXORdisk II or EXORdisk III
- MDOS 3.0 operating system

The user of additional memory or a line printer is optional.

1.3 TERMINAL PREPARATION

The heart of the EXORterm is the CRT terminal in whose housing the actual microcomputer, memory, interfaces, and other elements are incorporated. Considerable flexibility is designed into the CRT to accommodate a variety of different environments and configurations. Three groups of option switches on the back of the terminal housing itself must be set as follows for proper operation of the Editor.

GROUP ONE		GROUP TWO		GROUP THREE
ENABLE	ON	DUPLEX	FULL	BAUD RATE - SELECT ONLY ONE. 9600 is the most desirable.
DISPLAY	OFF	PARITY	NO	
TRANS MODE	OFF		EVEN	
VIDEO INV	OFF	XMIT WORDS	8-BIT	
A	ON	STOP BIT	1	
B	OFF	CONNECTION	DIRECT	
C	OFF	MODEM TYPE	103	
SPEC CHAR	OFF	TURN AROUND	8-CHAN	
LINE FREQ		CODE SEL	EDT	

1.4 LOADING THE DISK OPERATING SYSTEM (MDOS)

The actual procedure for initially loading the MDOS Disk Operating System will vary slightly between different hardware configurations. Refer to the EXORdisk II/III Operating System User's Guide for the exact initializing procedure for your hardware configuration. Typical procedures are:

EXORterm 200 (or equivalent TTY compatible terminal)	depress restart key EXBUG 1.2 MAID (no carriage return) *800;G (no carriage return) MDOS 3.0 =
EXORterm 220	depress restart key *E MDOS (carriage return) MDOS 3.0 =

The equal sign is an MDOS prompt character. MDOS commands may now be entered.

1.5 INVOKING THE EDITOR

The MDOS command E will invoke the Editor to create or modify a named file. The syntax for the edit command is as follows. All elements bounded by brackets are optional; e.g., [;options]. All elements outside the brackets must be entered. All user input is terminated by a carriage return (RETURN).

SYNTAX: E FILENAME1 [,[FILENAME2][,:DRIVE]][;[TS][FN][S]]

WHERE: FILENAME1 is the file to be edited. If FILENAME1 exists, it will be edited. If it does not exist, it will be created and made available for editing.

FILENAME2 is the file name of the edited result. FILENAME2 is only valid if FILENAME1 exists. If FILENAME2 is not specified, the original FILENAME1 will be deleted at the end of the edit process, and the edited result will be renamed FILENAME1. If FILENAME2 is specified, FILENAME1 is retained as an MDOS file without any alteration by the Editor.

DRIVE specifies the drive on which to put the second scratch file. The Editor will pick the optimum drive if DRIVE is not specified. The Editor uses temporary files named SCRATCH1 and SCRATCH2. SCRATCH1 is used whenever FILENAME1 exists. When the edit is terminated via QUIT or a SAVE command, SCRATCH1 is renamed FILENAME2 (or FILENAME1 if FILENAME2 was not specified). SCRATCH2 is used to support multiple edit passes in a single edit session.

TS specifies the initial tab stop settings for the Editor as follows:

- **A** implies assembler tabs (8,15,23) with the space bar as a special tab character and whereby an "*" in column 1 will disable the special tab character.
- **F** implies FORTRAN tabs (7) with the space bar as a special tab character and whereby a "C" in column 1 will disable the special tab character.
- **C** implies COBOL tabs (6,9,12) with the space bar as a special tab character.

FN is the file numbering option. An N implies edit with line numbers. A -N implies edit without line numbers. FN defaults to N when FILENAME1 does not exist, and is determined by the first line in the FILENAME1 when it does exist. Whenever a line numbered file is edited without using the line numbers, the line numbers are treated as data.

S specifies that the Editor is to be initialized in the scroll mode, and that the user is editing from a non-EXORterm type device. With this option, the user is restricted to scroll mode command editing.

The response to the E command is shown in the following example.

EXAMPLE:

```
=E LTR2 (carriage return)
MDOS EDITOR RELEASE X.X (X.X is the version number)
COPYRIGHT BY MOTOROLA 1978
```

When FILENAME1 exists, the Editor will check to determine if there is sufficient disk space to hold the edited result and the work file on the drives specified. If there is not sufficient space to store an edited file the size of which is identical to the input file, the Editor will display "INSUFFICIENT DISK SPACE ON DRIVE X CONTINUE (Y/N)?" A "Y" response will cause the Editor to continue; an "N" response will cause the Editor to return control to MDOS. Once the Editor has been invoked, it will display a message indicating that the file, FILENAME1, was opened or created new, and whether the file is line numbered (with line numbers).

1.6 MODES OF OPERATION

The Editor has two processing modes: CRT-MODE and SCROLL-MODE. The Editor is configured to come up in the CRT-MODE when it is loaded (unless the S option of the E command is used).

1.6.1 CRT-MODE

The CRT-MODE supports both page editing and command editing. When the Editor is used to edit an existing file, it dynamically adapts to command editing (refer to Figure 1-1). With command editing, the user enters edit commands at the cursor position following the prompt (>) on line 22, while continually viewing a portion of the file on the CRT screen (lines 1 through 20). Thus, the screen becomes a window where the portion of the file currently acted upon is displayed. The Editor prompt in this window (column 1) serves as a pointer to indicate the current file position. Table 1-1 provides a summary of the commands. Table 1-2 lists the special character commands that are controlled by the use of the terminal's special-purpose keys when performing command editing.

Page editing can only be initiated in the CRT-MODE. It is initiated when the Editor is invoked to create a new file (refer to Figure 1-2), or when one of the following commands is entered:

> CHANGE (without a STRING2)
> INSERT (for text files without line numbers)
> NUMBER (for source files with line numbers)

Page editing is terminated via a mode change request (use of the F1 or F2 function keys).

With page editing, the user is positioned on the window and uses the CRT control characters to insert, change, or delete characters and/or lines within the file. Table 1-3 lists the special character commands that are controlled by use of the terminal's special purpose keys when performing page editing.

The Editor will accept only valid printable ASCII characters. Any non-printable character received from the terminal or file in a data stream will be ignored. The EXORterm keyboard provides the means of inputting the full 128-character ASCII set, plus various function keys. The function keys F1 through F7 are used to select the special character commands in the CRT-MODE, as shown in Tables 1-1 and 1-2. In this mode, the bottom two lines on the screen are used to display the mnemonic function name for each of the seven function keys.

```
>0010  NAM POLL
 0020  OPT MEM
 0030 PIA1AC EQU $4005
 0040 PIA1BC EQU $4007
 0050 PIA2AC EQU $4009
 0060 PIA2BC EQU $400B
 0070  ORG $100
 0080 POLL LDA A PIA1AC
 0090  BMI ROUT1
 0100  ROL A
 0110  BMI ROUT2
 0120  LDA A PIA1BC
 0130  BMI ROUT3
 0140  ROL A
 0150  BMI ROUT4
 0160  LDA A PIA2AC
 0170  BMI ROUT5
 0180  ROL A
 0190  BMI ROUT6
 0200  LDA A PIA2BC
EDITING OLD FILE: LTR2    .SA:0 WITH LINE NUMBERS

F1      F2       F3       F4       F5       F6       F7
CRT     SCROLL   PAGE ^   PAGE v   LINE ^   LINE v   DUP  LTR1     .SA:0
```

FIGURE 1-1. CRT-MODE (Command Editing)

TABLE 1-1. Edit Commands

COMMAND	DESCRIPTION
C, CHANGE	Change a string within a record (or a group of records).
DEL, DELETE	Delete data from the file.
D, DUPLICATE	Copy a group of records from one place within a file to another place within the file.
E	Terminate the edit session. Begin editing another file.
EX, EXTEND	Insert data at the end of records.
F, FIND	Find a string or a line.
I, INSERT	Insert data into the file. (For editing files without line numbers.)
L, LIST	Display a portion of the file.
MERGE	Retrieve records from another MDOS file and include them in the edited file.
MOVE	Move a group of records from one place within the file to another place within the file.
N, NUMBER	Insert data into the file and prompt with the next line number. (For editing files with line numbers.)
PRINT	Print data at an associated line printer.
QUIT	Terminate the edit session and return control to MDOS.
R, RANGE	Establish default values for the vertical and horizontal ranges. (For LIST, PRINT, CHANGE, FIND, and SAVE.)
RESEQUENCE	Number, renumber, or unnumber the file.
SAVE	Save the edited file into the named file.
S, SEARCH	Find the next occurrence of a string.
TAB	Alter the tab settings used for data input.
V, VERIFY	Display at the terminal, any time a change is made to the file, the values after the change.
X, XTRACT	Insert the data in the buffer into the file.
1-9999	Insert, modify, or delete a line from the file. (This command type is for editing files with line numbers.)

TABLE 1-2. Command Editing (CRT-MODE)

COMMAND	DESCRIPTION
←	Move the cursor to the left one column with no character modification.
→	Move the cursor to the right one column with no character modification.
↑	Move the display pointer up one line on the display, with page wraparound.
↓	Move the display pointer down one line on the display, with page wraparound.
CTL-W	Pause the processing of this list.
CTL-X	Cancel this line.
|←	Move the cursor to the left to the previous tab stop.
→|	Move the cursor to the right to the next tab stop.
DEL	Delete the previous character.
BREAK	Stop processing; await a new command.
INS CHAR	All columns in the line, starting with the cursor column, are moved right one column. The cursor is then set at the inserted space.
DEL CHAR	All columns in the line, starting with the cursor column, are moved left one column. The cursor does not change position.
LINE ERASE	All data from the cursor position to the end of the line is deleted.
F2	Change the mode to SCROLL-MODE.
F3	Scroll the screen forward one page.
F4	Scroll the screen back one page.
F5	Scroll the screen forward one line.
F6	Scroll the screen back one line.

```
>0010









EDITING NEW FILE: LTR3    .SA:0 WITH LINE NUMBERS
>
F1      F2        F3        F4        F5        F6         F7
CRT     SCROLL    PAGE ^    PAGE ˇ    LINE ^    LINE ˇ     DUP   LTR3      .SA:0
```

FIGURE 1-2. CRT-MODE (Page Editing)

TABLE 1-3. Page Editing (CRT-MODE)

COMMAND	DESCRIPTION
←	Move the cursor to the left one column with no character modification.
→	Move the cursor to the right one column with no character modification.
↑	Move the display pointer and the cursor up one line on the display, with page wraparound.
↓	Move the display pointer and the cursor down one line on the display, with page wraparound.
CTL-X	Cancel this line.
|←	Move the cursor to the left to the previous tab stop.
→|	Move the cursor to the right to the next tab stop.
DEL	Delete the previous character.
INS CHAR	All columns in the line, starting with the cursor column, are moved right one column. The cursor is then set at the inserted space.
DEL CHAR	All columns in the line, starting with the cursor column, are moved left one column. The cursor does not change position.
INS LINE	Insert a line at the current file position.
DEL LINE	Delete the line at the current file position.
LINE ERASE	All data from the cursor position to the end of the line is deleted.
F1	Stop page editing; set the mode to the CRT-MODE.
F2	Stop page editing; set the mode to the SCROLL-MODE.
F3	Scroll the screen forward one page.
F4	Scroll the screen back one page.
F5	Scroll the screen forward one line.
F6	Scroll the screen back one line.
F7	Duplicate the data on the line immediately above this one.

1.6.2 SCROLL-MODE

In the SCROLL-MODE, the user performs command editing with each command line scrolling up and the resultant display on the next line. Table 1-4 lists the special character commands used in SCROLL-MODE. Figure 1-3 shows a typical program being edited in SCROLL-MODE.

TABLE 1-4. Command Editing (SCROLL-MODE)

COMMAND	DESCRIPTION
CTL-W	Pause the processing of this list.
CTL-X	Cancel this line.
CTL-D	Redisplay this line without character deletions.
→|	Echo spaces to move to the right to the next tab stop.
DEL	Delete the previous character and indicate so by echoing the deleted character.
BREAK	Stop processing; await a new command.
F1	Change the mode to CRT-MODE.

```
=E LTR3;S
MDOS EDITOR RELEASE 3.0
COPYRIGHT BY MOTOROLA 1978

EDITING OLD FILE: LTR3    .SA:0
>L 10-160
0010  NAM POLL
0020  OPT MEM
0030 PIA1AC EQU $3005
0040 PIA1BC EQU $4007
0050 PIA2AC EQU $4009
0060 PIA2BC EQU $400B
0070  ORG $100
0080 POLL LDA A PIA1AC
0090  BMI ROUT1
0100  ROL A
0110  BMI ROUT2
0120  LDA A PIA1BC
0130  BMI ROUT3
0140  ROL A
0150  BMI ROUT4
0160  LDA A PIA2AC

>
```

FIGURE 1-3. SCROLL-MODE

CHAPTER 2

EDIT COMMANDS

2.1 INTRODUCTION

There are two levels of edit features available: a basic set, which the user may master in a relatively short period of time, and an advanced set, which gives the user much more flexibility in editing. The advanced set may be mastered as needed.

The basic command set includes the 1-9999, CHANGE, DELETE, PRINT, SAVE, and QUIT commands, along with the elementary edit feature.

The Editor prompts the user for the next command by displaying a greater-than sign (>). The command is entered by typing a carriage return. Many of the commands have an abbreviated form, or alias, that the programmer may use if desired. A description of the Editor commands follows.

2.2 COMMAND SYNTAX

Editor commands may be entered in either upper or lower case and in either a short or long form. Only the first four characters of any command have significance, so that all commands may be abbreviated to the first four characters. In all of the commands, the character * may be used to indicate the current line position when referring to a line number.

In defining the commands provided by the Editor, the most common syntactical elements, vertical and horizontal range, are defined as follows:

```
VERTICAL-RANGE = [[BEGINNING-LINE-NUMBER] [- ENDING-LINE-NUMBER]]
                 OR
                 [[COUNT1 - ] COUNT2]
```

WHERE: BEGINNING-LINE-NUMBER / ENDING-LINE-NUMBER apply to line numbered files.
COUNT1 / COUNT2 apply to unlined files.
COUNT1 is the offset from the current line at which to begin the function performed. The offset may be negative.
COUNT2 is the number of lines upon which to perform the function.

The default for omitted values (where not overridden by the RANGE command) will result in the function being performed on the current line.

```
HORIZONTAL-RANGE = [:COLUMN1 [ - COLUMN2]] OR [:Fi [ - Fi+n]]
```

WHERE: COLUMN1 is the column on each line in which to begin the function to be performed.
COLUMN2 is the column on each line in which to end the function to be performed.
Fi is the field defined by the i'th tab stop through the i'th + 1 tab stop.

The default values where no horizontal range is specified will result in the function being performed on the entire line.

2.3 1-9999 COMMAND

FUNCTION: Insert, modify, or delete a line from the file. (This command applies to line-numbered files only.)

SYNTAX: LN DATA

WHERE: LN is the line number of the line to be inserted, modified, or deleted. DATA is the new value of the line.

The entry of a line number with no data will imply a request to delete the line.

CRT-MODE: The line number inserted, modified, or the line immediately following the line deleted is at the current line position.

EXAMPLES:

1. To insert line number 200 into the file.
   >200 DATA ON LINE 200cr

2. To change line number 200.
   >200 NEW DATA ON LINE 200cr

3. To delete line number 200 from the file.
   >200cr

4. To insert a null line numbered 200 (line 200 does not currently exist in the file).
   >200cr

2.4 CHANGE COMMAND

FUNCTION Change a string within a record (or a group of records).

SYNTAX: CHANGE [VERTICAL RANGE] [HORIZONTAL RANGE] [i.TRANSPARENT CHAR] [;COUNT] STRING] [STRING2] [REPETITION]

ALIAS: C,change,c

WHERE: STRING1 is the target string to be found and changed.
STRING2 is the value to which STRING1 is to be changed.
COUNT indicates that the COUNT'th occurrence of STRING1 in a line is the target string.
REPETITION indicates the number of times STRING1 is to be found and changed.
TRANSPARENT-CHAR is a character within STRING1 that is to be ignored when testing for equality to STRING1.

An "A" in the COUNT or REPETITION field indicates all occurrences. The default value for COUNT is 1 (the first occurrence), and the default value for REPETITION is all lines within the VERTICAL-RANGE. String data is delimited by the first non-blank, non-numeric character. The same delimiter must also terminate STRING1 and STRING2 (if present). Care should be used in picking a string delimiter as the use of "." or ";" when the TRANSPARENT-CHAR or COUNT is not specified will cause unpredictable results. When the change command is executed with no parameters, the next occurrence of the string data last specified by the user is modified. A null STRING1 will be interpreted to imply insert STRING2 at the beginning of the HORIZONTAL-RANGE. A null STRING2 will be interpreted to imply delete STRING1. If STRING2 does not exist and the user is in the CRT-MODE, the user will begin page editing with the cursor resting on the first character of STRING1. If STRING2 does not exist and the user is in the SCROLL-MODE, the target line will be displayed to STRING1 and the user is placed in the input process (see the INSERT and NUMBER commands). When only COLUMN1 (Fi) of the HORIZONTAL-RANGE is specified and no string is specified, the following will occur: if the user is in the CRT-MODE, the user will begin page editing with the cursor resting on the column position (or the field position) specified; if the user is in the SCROLL-MODE, the target line will be displayed to COLUMN1 (Fi) and the user placed in the input process (see the INSERT and NUMBER commands). A change command with only a null STRING1, in the CRT-MODE, initiates page editing at the current line position.

CRT-MODE: The last data line modified by the change is at the current position line.

EXAMPLES:

1. Change the string SAM to BILL on the current position line.
>C /SAM/BILL/cr

2. Change the first occurrence of the string SAM to BILL on each line from line numbers 100 through 200, inclusive.
>C 100-200 /SAM/BILL/cr

3. Change the first occurrence of the string SAM (that occurs after column 30 and before column 60) to the string BILL on line number 100.
>C100 :30-60 /SAM/BILL/cr

4. Change the third occurrence of the string SAM to BILL on the current position line.
>C ;3/SAM/BILL/cr

5. Change all occurrences of the string SAM to BILL on each of the lines from line number 100 to 200, inclusive.
>C 100-200 ;A/SAM/BILL/cr

6. Delete the string BOB from line number 120.
>C 120 /BOB//cr

7. Insert the string BOB at the beginning of the current line.
>C //BOB/cr

8. Change the strings LABELx (where x can be any character or digit) to the string 1.25 on the line numbers from 100 to 500, inclusive.
>C 100-500 .X/LABELX/1.25/cr

9. Change the first occurrence of the string SAM to BILL on the lines from line number 100 to 200. Then change the next occurrence of the string SAM to BILL from wherever it is to the end of the file.
>C 100-200 /SAM/BILL/1cr
>Ccr

10. Change the second occurrence of the string LABxL (where x can be any character or digit) to the string LABEL where it occurs between columns 30 and 60 on line numbers 100 to 130, inclusive.
>C 100-130 :30-60 .X;2/LABXL/LABEL/cr

11. Change the string SAM to BILL on the next 5 lines (where the file being edited does not have line numbers).
>C *-5 /SAM/BILL/cr

12. Begin page editing at the beginning of this line.
>C //cr

13. Begin page editing at the beginning of the string SAM on this line.
>C /SAM/cr

14. Begin page editing on column 12 of line number 100.
>C 100 :12cr

15. Begin page editing on the second occurrence of the string X on line 200.
>C 200 ;2/X/cr

2.5 DELETE COMMAND

FUNCTION: Delete data from the file.

SYNTAX: DELETE [VERTICAL-RANGE] [HORIZONTAL-RANGE]

ALIAS: DEL,delete,del

Specification of a HORIZONTAL-RANGE will cause the data within the range to be deleted from each line within the VERTICAL-RANGE.

CRT-MODE: The line immediately following the last line affected is at the current position line.

EXAMPLES: 1. Delete the line at the current file position.
>DELcr

2. Delete the lines between lines 100 and 200, inclusive.
>DEL 100-200cr

3. Delete the data between columns 10 and 20, inclusive, on line 210.
>DEL 210 :10-20cr

4. Delete line number 210.
>DEL 210cr

5. Delete the next 5 lines (when editing a file without line numbers).
>DEL *-5cr

2.6 PRINT COMMAND

FUNCTION: Print data at an associated line printer.

SYNTAX: PRINT [VERTICAL-RANGE] [SPACING]

ALIAS: Print

WHERE: SPACING is "D" to indicate double spacing and "T" to indicate triple spacing.

CRT-MODE: The print command does not alter the display.

EXAMPLES: 1. Print the entire file.
>PRINTcr

2. Print this line.
>PRINT *cr

3. Print the lines between line numbers 100 and 200, inclusive, with double spacing.
>PRINT 100-200 Dcr

4. Print the next 10 lines (when editing a file without line numbers) with triple spacing.
>PRINT *-10 Tcr

2.7 SAVE COMMAND

FUNCTION: Save the edited file into the named file.

SYNTAX: SAVE [[VERTICAL-RANGE] FILENAME]

ALIAS: Save

WHERE: FILENAME implies that an extract function is to be performed. The referenced data (as modified by the edit) is written into the new file: FILENAME.

FILENAME must not already exist as an MDOS file. If no VERTICAL-RANGE is specified, the current modified version of the file is written into FILENAME. When FILENAME is not specified, the current modified version of the source file will replace the old version, and the old version is deleted from MDOS. The new version is then reopened and repositioned to the beginning of the file for further editing.

CRT-MODE: The save command with FILENAME specified does not alter the display and, without FILENAME specified, resets the display to the beginning of the file.

EXAMPLES: 1. Save the edited file to disk.
>SAVEcr

2. Save the edited file into a new file named SAM.
>SAVE SAMcr

3. Extract the lines between line numbers 100 and 500, inclusive, and put them into a new file named SAM.
>SAVE 100-500 SAMcr

4. Extract the next 20 lines (when editing a file without line numbers) and put them into a new file named SAM.
>SAVE *-20 SAMcr

2.8 QUIT COMMAND

FUNCTION: Terminate the edit and return control to MDOS.

SYNTAX: QUIT [A]

ALIAS: Quit

WHERE: A implies abort the edit.

The quit command causes the results of the edit to be saved prior to termination unless the A option is specified.

CRT-MODE: The screen is cleared and the terminal is reset.

EXAMPLES: 1. Terminate the edit and save the results.
>QUITcr

2. Abort the edit and do not modify the source file.
>QUIT Acr

2.9 DUPLICATE COMMAND

FUNCTION: Copy a group of records from one place within a file to another place within the file.

SYNTAX: DUPLICATE [VERTICAL-RANGE] [/NEW-LINE-NUMBER [,NEW-INCREMENT]]

ALIAS: D,duplicate,d

WHERE: NEW-LINE-NUMBER is the destination line number for the first line copied (line numbered files only).
NEW-INCREMENT is the line number increment to be applied to each subsequent line of the copy.

The duplicate command will copy the requested data into a buffer area. The data can then be moved from the buffer via the XTRACT command. The duplicate command, when used with line numbered files specifying the NEW-LINE-NUMBER, will automatically invoke the XTRACT command to copy the data from the buffer to the target area. See the BLOCK/LINE MERGE RULES on the following page for the treatment of line number conflicts.

CRT-MODE: If the NEW-LINE-NUMBER was not utilized, the screen is unchanged. If it was specified, the current position line contains the line following the last line copied at its new location.

-- EXAMPLES: 1. Duplicate this line (put this line into the XTRACT buffer).
>Dcr

2. Duplicate lines 100 to 300, inclusive.
>D 100-200cr

3. Duplicate the next 10 lines (when editing a file without line numbers).
>D *-10cr

4. Duplicate this line and put it at line number 111.
>D /111cr

5. Duplicate line numbers 110 to 130, inclusive, and put them at line number 151, with an increment of 2 between lines.
>D 110-130/151,2

6. Duplicate line 130 and put it at 143.
>D 130/143cr

2.10 BLOCK/LINE MERGE RULES

When the establishment of new line numbers for a block insert generates a conflict (a potential duplication or overlap of line numbers), the editor will display:

"LINE NUMBER CONFLICT AT XXXX RESEQUENCE (Y/N)?"

If the user responds with a Y, the balance of the file will be resequenced only until there is no further conflict. If the user responds with an N, the duplicated lines in the file will be overlayed with the new lines. Consider the following example:

User file prior to any editing:

```
010 1st line
020 2nd line
030 3rd line
040 4th line
050 5th line
060 6th line
070 7th line
080 8th line
090 9th line
100 10th line
```

User enters: DUPLICATE 10-50/74,4

The Editor will begin processing:

```
010 1st line
020 2nd line
030 3rd line
040 4th line
050 5th line
060 6th line
070 7th line
074 1st line
078 2nd line
```

LINE NUMBER CONFLICT AT 80 RESEQUENCE (Y,N)?

If the user types Y, the rest of the file will be:

```
082 3rd line
086 4th line
090 5th line
091 8th line
092 9th line
100 10th line
```

If the user types N, the rest of the file will be:

```
080 8th line
082 3rd line
086 4th line
090 5th line
100 10th line
```

2.11 E COMMAND

FUNCTION: An Editor command that will terminate this edit session and initiate the editing of another named file. (For command specifics, see the INVOKING THE EDITOR section.)

2.12 EXTEND COMMAND

FUNCTION: Insert data at the end of records.

SYNTAX: EXTEND [VERTICAL-RANGE] STRING

ALIAS: EX,extend,e

WHERE: STRING is the value to be appended to the end of the records in the VERTICAL-RANGE.

The STRING must be delimited by the first non-blank non-numeric character. The same delimiter must be used to terminate the string.

CRT-MODE: The current position line is the last line modified by the Extend command.

EXAMPLES: 1. Put a comma at the end of the line at the current file position.
>EX /,/cr

2. Put a period at the end of line 100.
>EX /./cr

3. Put a period at the end of all lines between line numbers 100 and 200, inclusive.
>EX 100-200 /./cr

2.13 FIND COMMAND

FUNCTION: Find a string or a line.

SYNTAX: FIND [VERTICAL-RANGE] [HORIZONTAL-RANGE] [.TRANSPARENT-CHAR] [;COUNT] STRING [REPETITION]]

ALIAS: F,find,f

WHERE: TRANSPARENT-CHAR is a character within the STRING that is to be ignored in determining search satisfaction.
STRING is the string to be found.
COUNT indicates that the COUNT'th occurrence of the STRING in a line is the target string.
REPETITION indicates the number of times the STRING is to be found.

The STRING must be delimited by the first non-blank, non-numeric character. The same delimiter must be used to terminate the string. When a Find command is executed with no parameters, it implies a request to find the next occurrence of the last STRING entered by the Find command. Used in this fashion, the Find command begins the string search at the current position line. When the Find command is executed with only a VERTICAL-RANGE specified, it implies a request to position the Editor at the start of the VERTICAL-RANGE. FIND 0 will imply a request to position the editor at the beginning of the file and, conversely, FIND 9999 will imply a request to position the Editor at the end of the file. When the VERTICAL-RANGE is not specified, the string search will begin at the beginning of the file. An A in the COUNT or REPETITION field will imply a request to find all the occurrences of the STRING. The default value for COUNT is 1, and the default value for REPETITION is 1. When the REPETITION is greater than 1, the screen, in the CRT-MODE, will display only the lines with an occurrence of the STRING. If the screen gets full, the display will scroll up until the list is exhausted or the user types CTL-W to pause the list or BREAK to abort the list. The screen may be returned to the normal CRT-MODE display by the entering of a null command or any other valid Editor command.

CRT-MODE: The current position line will always be the line with the last occurrence of the STRING.

EXAMPLES: 1. Position the Editor to the beginning of the file.
>F 0cr

2. Position the Editor to the end of the file.
>F 9999cr

3. Position the Editor to line number 140.
>F 140cr

4. Position the Editor on the first occurrence of the string SAM in the file.
>F /SAM/cr

5. Position the Editor on the first line with two occurrences of the string SAM between line number 100 and 200.
>F 100-200 ;2/SAM/cr

6. List all occurrences of the string SAM.
>F /SAM/Acr

7. List all lines with 2 or more occurrences of the string SAM.
>F ;2/SAM/Acr

2.14 INSERT COMMAND

FUNCTION: Insert data into the file. (For editing files without line numbers)

SYNTAX: INSERT [OFFSET]

ALIAS: I,insert,i

WHERE: OFFSET is the offset from the current position line at which to begin inserting.

The Insert command initiates the input process in which all data following the Insert command is inserted into the file until a null line is encountered (SCROLL-MODE) or mode-change is input (F1 or F2 keys).

CRT-MODE: When the Insert command is received, the current line position is set to the point at which input is to begin. The current line is cleared, pushing down all lines on the screen to make room for the data to be inserted. The cursor is then positioned on the cleared line to accept input. When page editing is terminated, the data below the current line is scrolled up, returning the current line position to the line it was positioned on prior to the insert. At that time, the cursor is returned to the command line. The Insert command in the CRT-MODE initiates page editing.

EXAMPLE: 1. To insert data into the file at the current file position.
>Icr

2.15 LIST COMMAND

FUNCTION: Display a portion of the file.

SYNTAX: LIST [VERTICAL-RANGE]

ALIAS: L,list,l

The List command will cause the data within the VERTICAL-RANGE to be displayed on the screen. If the list is too long for the screen, the display will be scrolled up until the list is exhausted or the user types CTL-W to pause the list or BREAK to abort the list. When the listing is finished or aborted, the user can return the screen to the CRT-MODE by entering a null command or any valid Editor command.

EXAMPLES: 1. Display the entire file.
>Lcr

2. Display only the current file position.
>L *cr

3. Display the lines between line numbers 100 and 200, inclusive.
>L 100-200cr

4. Display the next 10 lines (when editing a file without line numbers).
>L *-10cr

2.16 MERGE COMMAND

FUNCTION: Retrieve records from another MDOS file and include them in the edited file.

SYNTAX: MERGE [VERTICAL-RANGE] FILENAME [NEW-LINE-NUMBER [,NEW-INCREMENT]]

ALIAS: Merge

WHERE: FILENAME specifies a valid MDOS file.
VERTICAL-RANGE specifies the range in FILENAME to be copied into the edited file.
NEW-LINE-NUMBER is the line number to be assigned to the first line from FILENAME when it is inserted into the file.
NEW-INCREMENT is the increment to be added to each successive line number as the lines are inserted into the edited file.

If the VERTICAL-RANGE is not specified, it will imply the request to get the entire FILENAME. If the user is editing without line numbers, the data from FILENAME will be inserted at the current position line. The line numbers that may be in FILENAME are lost upon extract. The line numbers applied will be as specified via NEW-LINE-NUMBER and NEW-INCREMENT, if specified, or will default to the current position line plus 10 with an increment of 10. If the establishment of new line numbers causes a conflict, the BLOCK/LINE MERGE RULES shown following DUPLICATE command description will apply.

CRT-MODE: The current position line following a merge will be the line following the last line inserted by the merge.

EXAMPLES: 1. Insert all the lines in the file named SAM into the edited file at the current file location.
>MERGE SAMcr

2. Insert all the lines in the file named SAM into the edited file beginning with line number 400 with an increment of 1 between line numbers.
>MERGE SAM 400,1cr

3. Insert line numbers 100 to 300, inclusive, in the file named SAM into the edited file, beginning with line number 500 and with increments of 2 between lines.
>MERGE 100-300 SAM 500,2cr

2.17 MOVE COMMAND

FUNCTION: Move a group of records from one place within the file to another place within the file.

SYNTAX: MOVE [VERTICAL-RANGE] [/NEW-LINE-NUMBER [,NEW-INCREMENT]]

ALIAS: Move

WHERE: NEW-LINE-NUMBER is the destination line number for the first line moved (line numbered files only).
NEW-INCREMENT is the line number increment to be applied to each subsequent line of the move.

The Move command generates a DUPLICATE command followed by a DELETE command. For additional information, see those command descriptions.

EXAMPLES: 1. Move this line to the XTRACT buffer.
>MOVEcr

2. Move line number 100 to the XTRACT buffer.
>MOVE 100cr

3. Move the lines between line numbers 100 and 200, inclusive, into the XTRACT buffer.
>MOVE 100-200cr

4. Move the next 10 lines into the XTRACT buffer (when editing a file without line numbers).
>MOVE *-10cr

5. Move the line at the current file position to line number 131.
>MOVE /131cr

6. Move line 120 to 123.
>MOVE 120/123cr

7. Move the lines between 150 and 200, inclusive, to line number 190, with an increment of 1 between lines.
>MOVE 150-200/190,1cr

2.18 NUMBER COMMAND

FUNCTION: Insert data into the file and prompt user with the next line number for insert (for editing line numbered files only).

SYNTAX: NUMBER [NEW-LINE-NUMBER [,NEW-INCREMENT]]

ALIAS: N,number,n

WHERE: NEW-LINE-NUMBER is the first line number to be prompted.
NEW-INCREMENT is the value to be added to the NEW-LINE-NUMBER to form each succeeding line number prompted.

The Number command initiates the input process in which all data following the command is inserted into the file. The Number command with no parameters will cause the Editor to form the line number prompt by adding the default increment of 10 to the last line number in the file. Prompting will continue after each carriage return until: a null line is encountered (SCROLL-MODE) or a mode-change is encountered (F1 and F2 keys). At that time, the Editor will again accept edit commands.

CRT-MODE: When the Number command is received, the current position line is set to the point at which input is to begin. The current line is cleared, pushing down all lines on the screen to make room for the data to be inserted. The cursor is then positioned on the cleared line following the line number prompt. When page editing is terminated, the data below the current line is scrolled up, returning the current position line to the line following the last line input. At that time, the cursor is returned to the command line.

EXAMPLES:

1. Begin page editing and begin prompting line numbers at the end of the file in increments of 10.
   >Ncr

2. Begin page editing at line number 100.
   >N 100cr

3. Begin page editing at line number 200, with line number prompts in increments of 1.
   >N 200,1cr

2.19 RANGE COMMAND

FUNCTION: Establish a default value for the vertical and horizontal ranges.

SYNTAX: RANGE [VERTICAL-RANGE] [HORIZONTAL-RANGE]

ALIAS: R,range,r

The Range command with no parameters deletes the previously established ranges. The Range command provides the default ranges for LIST, PRINT, CHANGE, FIND, and SAVE.

EXAMPLES:

1. Set the default VERTICAL-RANGE to lines 100 to 300, inclusive.
   >R 100-300cr

2. Set the default HORIZONTAL-RANGE to edit a full 132 character line.
   >R :1-132cr

3. Set the VERTICAL-RANGE default to lines 100 to 200, inclusive, and set the HORIZONTAL-RANGE to columns 16 to 80, inclusive.
   >R 100-200 :16-80cr

2.20 RESEQUENCE COMMAND

FUNCTION: Number, renumber, or unnumber the file.

SYNTAX: RESEQUENCE [BEGINNING-LINE-NUMBER] [/NEW-LINE-NUMBER [,NEW-INCREMENT]] [N]

ALIAS: Resequence

WHERE: BEGINNING-LINE-NUMBER is the line number at which to begin resequencing.
NEW-LINE-NUMBER is the line number to be assigned to the first line resequenced.
NEW-INCREMENT is the increment to be applied to form each subsequent line number within the file.
N implies a request to strip the line numbers from the file.

If BEGINNING-LINE-NUMBER is not specified, the entire file will be resequenced. If NEW-LINE-NUMBER is not specified, resequencing will begin with the BEGINNING-LINE-NUMBER plus NEW-INCREMENT. If NEW-INCREMENT is not specified, it defaults to 10. The N option is mutually exclusive to the other options on this command.

CRT-MODE: The current line position is not altered by this command; however, the display will be altered to show the effect of the resequence.

EXAMPLES: 1. Resequence the entire file. Begin the new line numbers with line number 10, with increments of 10 between line numbers.
>RESEcr

2. Resequence the file from line number 150 to the end of the file. Use increments of 10 between line numbers.
>RESE 150cr

3. Resequence the entire file. Begin the line numbers with line number 100, with an increment of 2 between line numbers.
>RESE /100,2cr

2.21 SEARCH COMMAND

FUNCTION: Find the next occurrence of a string.

SYNTAX: SEARCH [STRING]

ALIAS: S,search,s

WHERE: STRING is the string to be found.

The STRING is delimited by the first non-blank character. The same delimiter must be used to terminate the string. The Search command will begin a string search at the line following the current line position. A Search command with no STRING will search for the next occurrence of the last string specified by a Search command or a Find command. Thus, the Search command is equivalent to a FIND 1-9999 /STRING/ in an unlined file, or FIND n-9999 /STRING/ in a numbered file (where n is the current line number + 1).

CRT-MODE: The Search command places the current line position at the next occurrence of the string.

EXAMPLES: 1. Find the next occurrence of the string SAM.
>S /SAM/cr

2. Find the first occurrence of the string BILL and then find the next occurrence.
>F /BILL/cr
>Scr

2.22 TAB COMMAND

FUNCTION: Alter the tab settings used for data input.

SYNTAX: TAB [CHAR] [COUNTN]1-20

ALIAS: Tab

WHERE: CHAR is the non-numeric, non-space character to designate a tab request during page editing.
COUNTN is the column position to which a tab is to be set.

When no CHAR is specified, the default tab key is the TAB key. The tabs can be reset by entering the Tab command with no parameters. The 1 to 20 COUNTN's specified must be separated by commas. The COUNTN values will add additional tab stops until the tabs are reset by a Tab command with no parameters.

CRT-MODE: The display is unchanged by the Tab command.

EXAMPLES: 1. Set tabs at columns 10 and 20 and use the character Z as a special tab key.
>TAB Z 10,20cr

2. Add an additional tab stop at column 15.
>TAB 15cr

3. Delete all tab stops.
>TABcr

2.23 VERIFY COMMAND

FUNCTION: Display at the terminal, any time a change is made to the file, the value after the change.

SYNTAX: VERIFY [OFF]

ALIAS: V,verify,v

WHERE: OFF disables the verify.

The verify is enabled by entering the Verify command with no parameters. With the verify enabled in the CRT-MODE, certain multiple update commands will display a list of occurrences rather than a sequential portion of the file surrounding the changed record. Should the list become too long for a single screen, the display will scroll up until the list is exhausted or the user types CTL-W to pause the list or BREAK to abort the list. The screen can be returned to the normal display by entering a null command or any other valid Editor command. Use of the BREAK key will cause the update command to stop processing as of the last line listed on the screen.

CRT-MODE: Execution of the Verify command does not alter the display. The Verify command causes other update commands to display as noted above.

EXAMPLES: 1. Enable verify.
>Vcr

2. Disable verify.
>V OFFcr

2.24 XTRACT COMMAND

FUNCTION: Insert the data in the buffer into the file.

SYNTAX: XTRACT [NEW LINE NUMBER [,NEW INCREMENT]] [A]

ALIAS: X,xtract,x

WHERE: NEW-LINE-NUMBER is the point at which to insert the first line from the buffer.
NEW-INCREMENT is the increment to be applied to each subsequent line number inserted into the file.
A indicates that the xtract buffer is to be deleted.

In unlined files, the xtract will always copy the data in the buffer into the file at the current position line. If the establishment of new line numbers generates a conflict, the BLOCK/LINE MERGE RULES (shown following the DUPLICATE command description) will be applied to resolve the conflict.

EXAMPLES: 1. Insert the data in the XTRACT buffer into the file at the current file position.
>Xcr

2. Insert the data in the XTRACT buffer into the file beginning with line number 200, with an increment of 10 between lines.
>X [illegible]

TABLE 1-5. Operator Messages

MESSAGE	PROBABLE CAUSE
EDITOR OVERLAY NOT AVAILABLE	All Editor overlay files must be in drive Ø throughout the edit session.
NAME REQUIRED	The Editor was invoked without specifying the file to be edited.
ILLEGAL FILE NAME	The name of the file specified is not a valid MDOS file name.
INVALID DEVICE	The device specified is not a valid MDOS device.
INVALID DRIVE	The drive number specified is not supported by MDOS.
XXXXXXXX.XX DUPLICATE FILE NAME	The file name specified cannot be created as it already exists.
XXXXXXXX.XX IS WRITE PROTECTED	The file name specified has been marked write protected and cannot be edited without specifying FILENAME2.
XXXXXXXX.XX IS DELETE PROTECTED	The file name specified has been marked delete protected and cannot be edited without specifying FILENAME2.
XXXXXXXX.XX HAS INVALID FILE TYPE	The file name specified is not an ASCII file and [illegible] edited.
OPTION CONFLICT	

TABLE 1-5. Operator Messages (cont'd)

COMMAND	PROBABLE CAUSE
SOURCE FILE SEQUENCE ERROR. RESEQUENCE REQUIRED	
	The source file contains non-numeric data in columns 1-4, or does not contain a space in column 5; or the line numbers in the source file are not in ascending order; or the line numbers in the source file are not unique.
WHAT?	An invalid edit command was entered.
COMMAND SYNTAX ERROR	
	A valid edit command was entered but the syntax of the command was not correct.
DEVICE NOT READY	
	A MERGE or PRINT command was received and the appropriate device is not ready.
XXXXXXXX.XX DOES NOT EXIST	
	A MERGE command was received but the specified FILENAME does not exist.
COMMAND ABORTED	
	The BREAK key was repressed during file access, and the command last entered has been aborted prior to completion.
STRING NOT FOUND	
	The string specified in the CHANGE, FIND, or SEARCH command cannot be found in the specified ranges.
INVALID FIELD	
	The field specified within the HORIZONTAL-RANGE has not been defined via tab stops.
LINE NUMBER OVERFLOW RESEQUENCE	
	An attempt to generate a line number resulted in a value greater than 9999, and wrapped around to or beyond 0000. The file must be resequenced or the command results are unpredictable.
XTRACT BUFFER OVERFLOW	
	An attempt to move or duplicate too large a block has occurred.
SOURCE LINES NOT FOUND	
	An attempt to MERGE, MOVE, DUPLICATE, or XTRACT occurred, and the referenced data does not exist; or the xtract buffer is empty.
EOF ENCOUNTERED	
	An attempt to MOVE or DUPLICATE a block of data was terminated by the end of a file. Data up to that point has been processed.

TABLE 1-5. Operator Messages (cont'd)

COMMAND	PROBABLE CAUSE
BOF OR EOF ENCOUNTERED	An attempt to page or scroll beyond the beginning or the end of the file has occurred.
LINE NUMBER CONFLICT AT XXXX RESEQUENCE (Y/N)?	Block editing has resulted in a line number conflict. The Editor will resolve the conflict according to the rules described in paragraph 1.6.9).
IS THE SOURCE FILE LINE NUMBERED (Y/N)?	A MERGE command was entered and the Editor needs to know whether or not to strip the first five characters of each record extracted.

EUROPEAN MOTOROLA SEMICONDUCTOR SALES OFFICES

HOLLAND
Motorola B.V.
[illegible]
Tel. [illegible]

SOUTH AFRICA
Motorola South Africa (Pty) Ltd
[illegible]

SWEDEN
Motorola AB
[illegible]

UNITED KINGDOM
Motorola Ltd
[illegible]
Sales Office
[illegible] Street, Television House
[illegible] Lancs
Sales Office
[illegible] Road, [illegible] Estate
[illegible] Scotland

HEADQUARTERS EUROPEAN OPERATIONS

SWITZERLAND
Motorola Inc
Semiconductor Products Group
[illegible] P.O. Box 8
[illegible]

FRANCHISED MOTOROLA SEMICONDUCTOR DISTRIBUTORS

HOLLAND
[illegible] 22 – Utrecht
[illegible]
[illegible] P.O. Box 25

HUNGARY
[illegible] P.O. Box 164

ITALY
[illegible]

SPAIN
[illegible]

SWEDEN
[illegible]

SWITZERLAND
[illegible]

TURKEY
[illegible]

UNITED KINGDOM
[illegible]

YUGOSLAVIA
[illegible]

EUROPEAN SEMICONDUCTOR FACTORIES

GERMANY
[illegible]

UNITED KINGDOM
[illegible]

Printed in Switzerland